Supplementary Information

From Codes to Causes: A Causal Interpretability Audit of Quantized Delta Representations in a Transformer World Model

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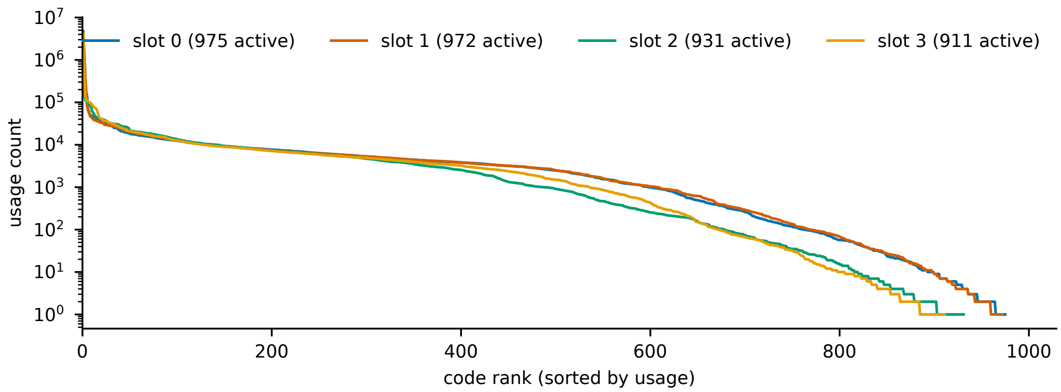
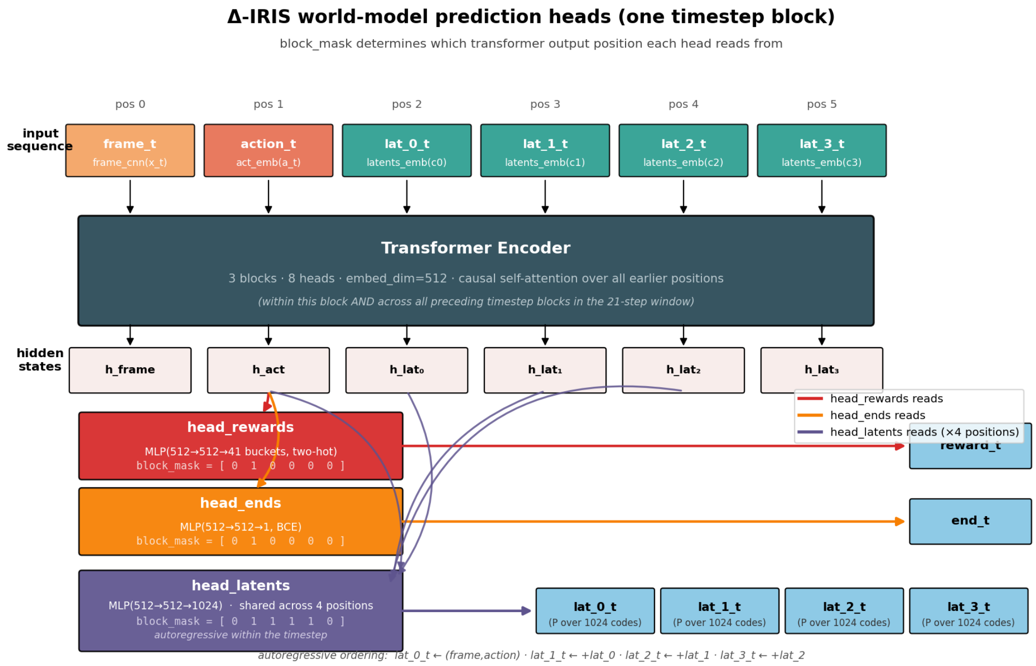
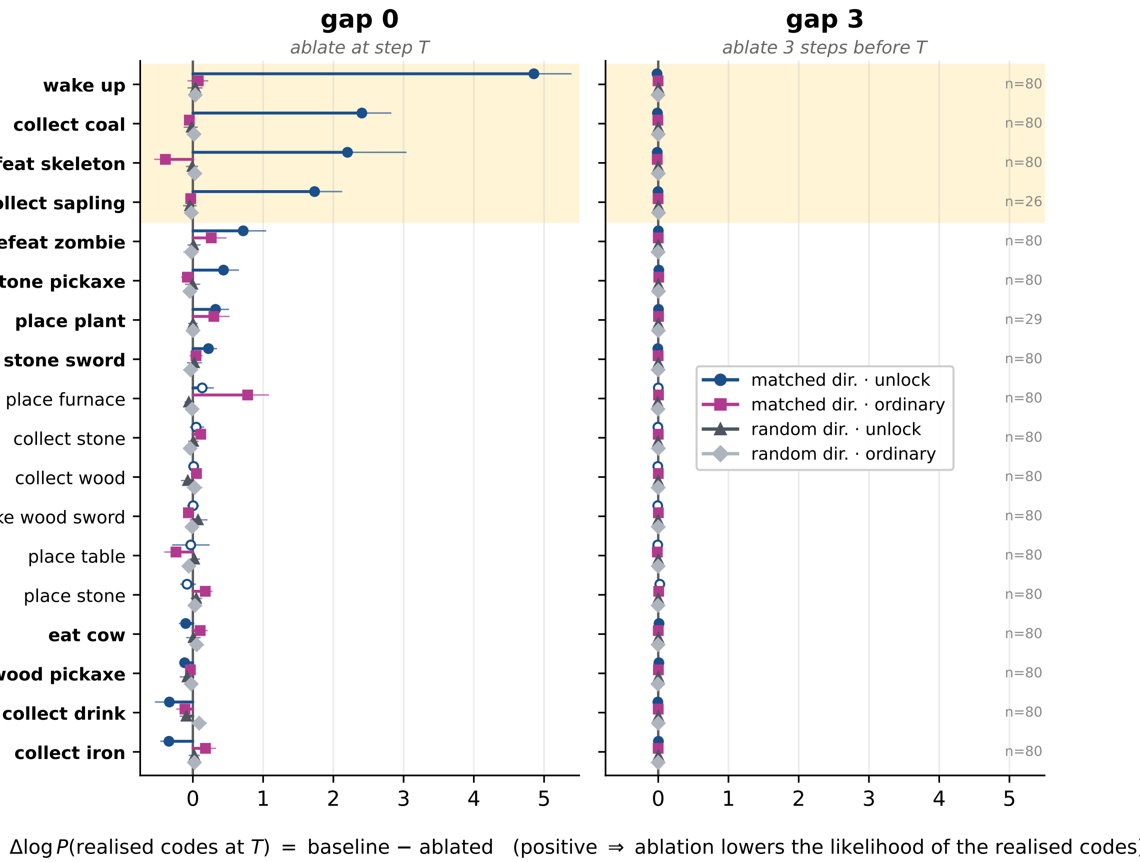
This document collects the displays deferred from the main text: codebook utilisation (Fig. S1), the prediction-head read positions (Fig. S2), single-step ablation (Fig. S3), decoded-frame filmstrips (Fig. S4) the partially observed (no-HUD) control (Fig. S5) and the behavioural cascade test (Fig. S6); and Tables S1–S8: detector codes, cross-corroborated dictionary features, matched-protocol reconstruction, HUD-shortcut margins, the sparse-autoencoder robustness sweep, the empirical random-direction null, steering, and direct logit attribution. All re-use the single frozen agent and the saved rollout corpora described in Methods.



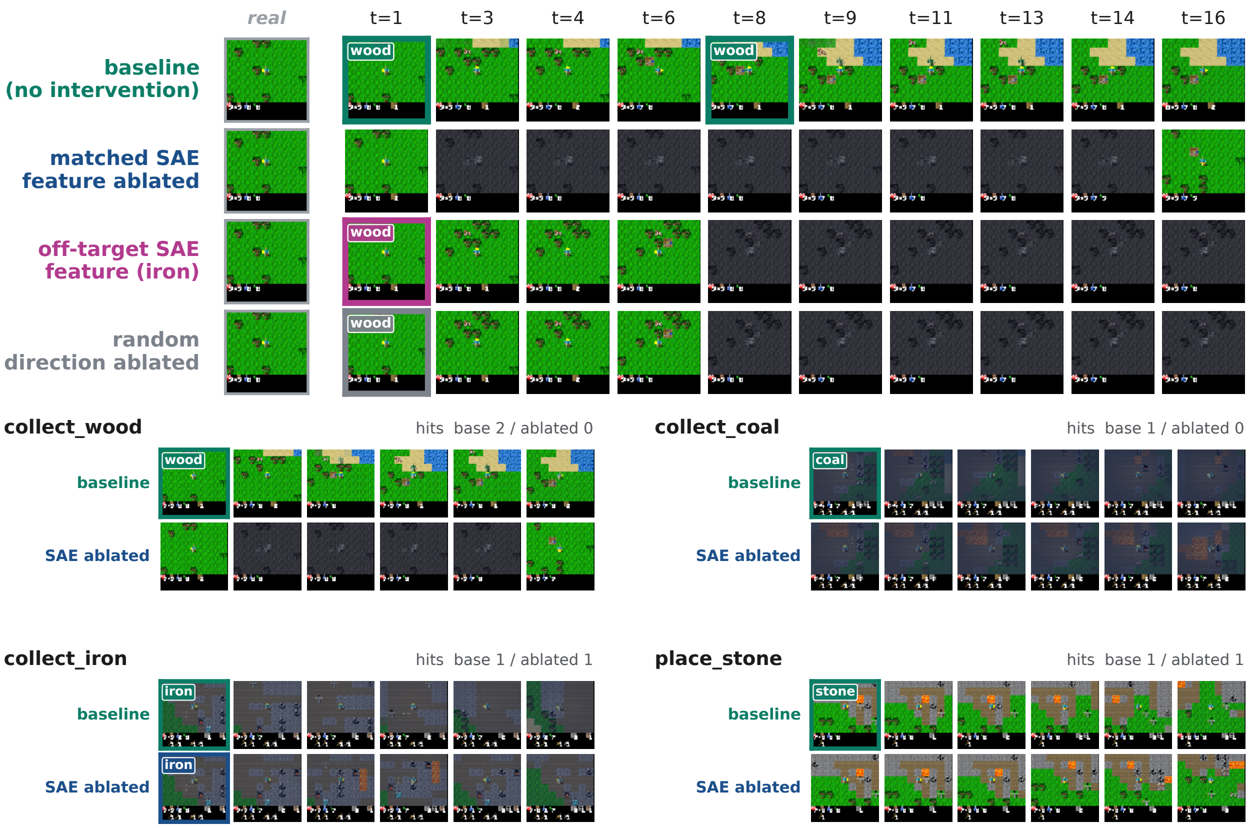
Supplementary Figure S1: **Codebook utilisation over the full 10^7-transition replay buffer.** Per-slot code-usage counts, sorted descending (log scale). All 1024 codes are active in the union of slots, but usage is Zipf-like: the head codes encode “no change in this quadrant” and the event vocabulary occupies the tail.



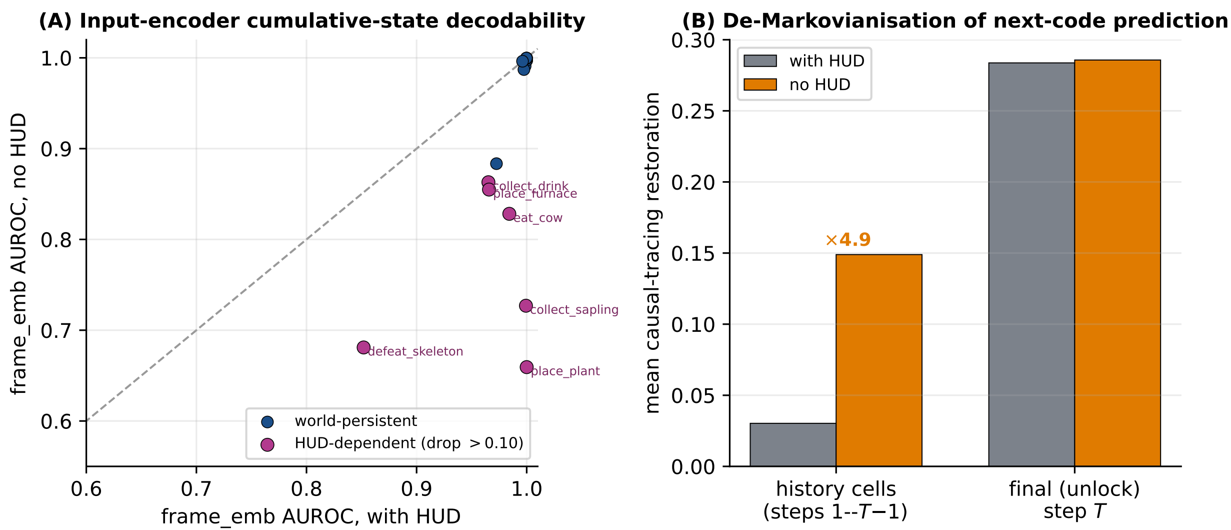
Supplementary Figure S2: **Detailed view of the Δ-IRIS prediction heads** (architecture follows Micheli et al.; our redrawing). `head_rewards` and `head_ends` read the post-transformer activation at the action position (block mask $[0, 1, 0, 0, 0, 0]$); `head_latents` reads positions 1–4 (mask $[0, 1, 1, 1, 1, 0]$) and predicts the four latent codes autoregressively within the timestep. The final latent position feeds subsequent attention but carries no prediction head.



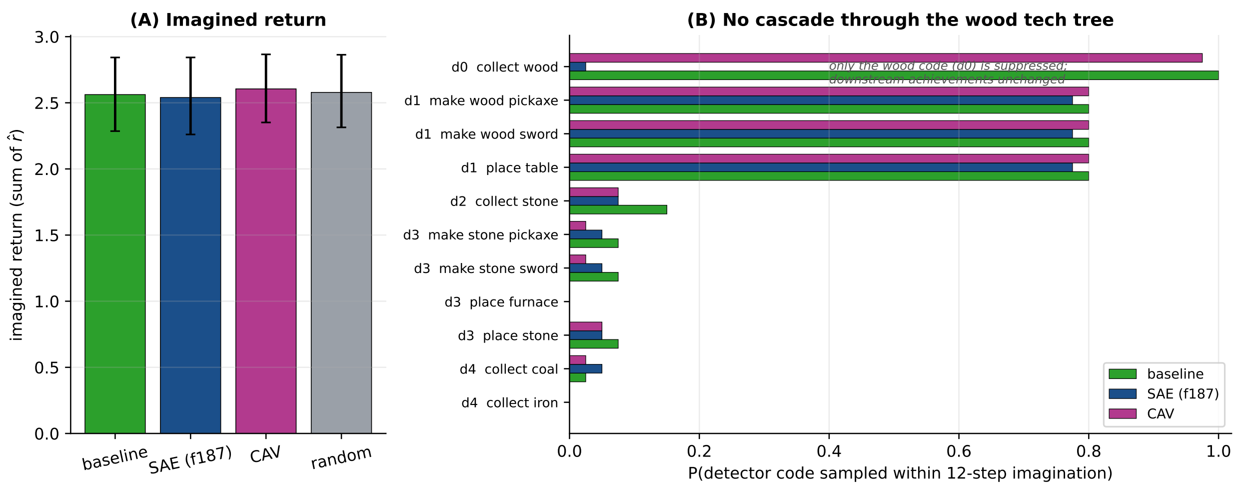
Supplementary Figure S3: **Single-step causal ablation.** $\Delta \log P$ = baseline − ablated log-likelihood of the four realised codes at the target step (positive ⇒ ablation lowered their likelihood) when a direction is projected out of the block-1 residual stream, at the target step (gap 0, left) or three steps earlier (gap 3, right), on one shared horizontal scale. Markers are means; whiskers 95% bootstrap CIs; the full 2×2 of {matched, random} direction × {unlock, ordinary} moment is shown. Random directions are inert; the matched direction is selective to unlock moments only for the dominant concepts; effects vanish entirely at gap 3 (the interventional face of the HUD shortcut).



Supplementary Figure S4: **Imagined trajectories under ablation (decoded frames).** Top: paired trajectories for `collect_wood` from a shared burn-in and sampling seed—no intervention; matched SAE feature ablated (wood never appears, dream degrades); off-target SAE feature ablated (the `collect_iron` feature, a magnitude-comparable control); random unit direction. Only the matched direction erases the event. Bottom: baseline vs matched-SAE ablation for four concepts, ranging from clean erasure (`collect_wood`) through partial suppression (`collect_coal`) to none (`collect_iron`, `place_stone`)—honest negatives whose best SAE features are not causally load-bearing.



Supplementary Figure S5: **Partially observed (no-HUD) control: with-HUD vs no-HUD.** A second agent (identical code, hyperparameters and seed) was trained on a Crafter variant that zeroes the on-screen inventory/vitals region; mean return falls from 16.2 to 9.7. **(A)** input-encoder (frame_emb) held-out AUROC for each cumulative-state concept, with HUD (x) vs no-HUD (y). World-persistent achievements (blue; placed objects, gathered materials, tools) stay near 1.0 because they leave visible marks in the scene, so the input-encoder shortcut survives; only the six genuinely HUD-only concepts (magenta) drop below the diagonal. A transformer block out-reads frame_emb on only 1 of 17 concepts, so the hidden state is not made linearly decodable from the blocks. **(B)** causal-tracing restoration mass on history cells (steps 1 to $T-1$) vs the final unlock step: with state hidden, mean historical restoration rises $\approx 5\times$ ($0.03 \rightarrow 0.15$) while the final-step mass is unchanged—next-code prediction de-Markovianises.



Supplementary Figure S6: **Behavioural cascade test: a residual-stream lesion of the `collect_wood` feature (f_{187}) does not change the agent's imagined behaviour.** Closed-loop imagination from `collect_wood` unlock moments (n=40; the actor–critic acts on each decoded frame), under baseline / SAE-f_{187} / wood-CAV / random projection at block 1. **(A)** imagined return (sum of the reward head) is unchanged across conditions (2.56, 2.54, 2.61, 2.58). **(B)** per wood-tree achievement, ordered by tech-tree depth from `collect_wood`, the probability its detector code is sampled within the 12-step horizon. Ablating f_{187} erases only the wood code itself (depth 0, 1.00 $\rightarrow$ 0.03) and does *not* cascade: the immediately downstream wooden-tool achievements (depth 1; sampled via the shared polysemantic code (s_3, c_{36})) and the stone tier are unchanged, and the CAV and random directions are equally inert. The SAE direction is causally load-bearing for the next-code distribution but behaviourally inert—it does not propagate through the tech tree—corroborating, with a ground-truth causal graph, that the read–write effect is confined to the code channel.

Supplementary Table S1: **Strongest detector code per achievement** (selection). n_{co}: co-occurrences of the code with the just-unlocked event; n_c: total occurrences; lift $= P(u \mid c)/P(u)$. All p-values are below 10^{-40} (uncorrected exact binomial); the full 431-cell table ships with the code release.

Achievement	Code	n_{co}/n_c	$P(u \mid c)$	Lift
`place_plant`	(s_3, c_{938})	477/519	0.92	543×
`collect_sapling`	(s_2, c_{707})	419/463	0.90	535×
`collect_iron`	(s_3, c_{75})	360/410	0.88	708×
`collect_coal`	(s_3, c_{578})	450/626	0.72	467×
`make_wood_pickaxe`	(s_3, c_{903})	19/35	0.54	325×
`place_table`	(s_3, c_{36})	480/1507	0.32	188×
`eat_cow`	(s_2, c_{648})	251/1127	0.22	134×
`place_furnace`	(s_0, c_{51})	89/412	0.22	132×
`collect_wood`	(s_3, c_{940})	474/2471	0.19	113×
`defeat_zombie`	(s_1, c_{727})	68/365	0.19	113×

Supplementary Table S2: **Cross-corroborated SAE features.** The feature's top achievement (lift within its top activation quartile), most-aligned CAV, and the (slot, code) with the highest conditional mean activation. The first five satisfy strict triple-corroboration ($|\cos| > 0.4$ to the same-named CAV); f_{372} is the third member of the (s_3, c_{36}) split.

Feature	Achievement (lift)	CAV (cos)	Code
f_{1742}	`collect_coal` (589×)	just[`collect_coal`] (+0.58)	(s_3, c_{578})
f_{1820}	`collect_iron` (624×)	just[`collect_iron`] (+0.41)	(s_3, c_{75})
f_{872}	`make_stone_pickaxe` (522×)	just[`make_stone_pickaxe`] (+0.48)	(s_3, c_{616})
f_{677}	`place_table` (527×)	just[`place_table`] (+0.45)	(s_3, c_{36})
f_{1451}	`make_wood_pickaxe` (548×)	just[`make_wood_pickaxe`] (+0.46)	(s_3, c_{36})
f_{372}	`make_wood_sword` (444×)	just[`make_wood_sword`] (+0.23)	(s_3, c_{36})

Supplementary Table S3: **Matched-protocol reconstruction** on a common 8,192-transition held-out batch. Perplexity is exp(entropy) of the token histogram over all slots. Δ-IRIS conditions its decoder on (x_t, a_t); the baselines reconstruct x_{t+1} from codes alone, so Δ-IRIS's lower error is expected and not a like-for-like comparison.

Tokenizer	MSE ↓	PSNR ↑	LPIPS ↓	perplexity	tokens/frame
Δ-IRIS (delta + action)	0.00021	36.7	0.048	80.7	4
frame-only ablation	0.0145	18.4	0.467	48.5	4
original IRIS	0.0056	22.5	0.080	109.2	16

Supplementary Table S4: **HUD-shortcut margins with bootstrap CIs.** For each cumulative-state concept, frame_emb held-out AUROC and the *worst* (smallest) AUROC difference against any transformer block, with a paired-bootstrap 95% CI (B=1000). frame_emb is not significantly below any block on 16 of 18 concepts; the two exceptions (`collect_sapling`, `place_plant`) trail by $< 10^{-3}$, with all representations above 0.999.

Concept	frame_emb AUROC	worst ΔAUROC	95% CI	not below
`collect_coal`	1.000	+0.046	[+0.043, +0.049]	yes
`collect_drink`	0.965	+0.046	[+0.040, +0.051]	yes
`collect_iron`	1.000	+0.065	[+0.061, +0.068]	yes
`collect_sapling`	0.999	−0.0007	[−0.0010, −0.0004]	no
`collect_stone`	0.999	+0.014	[+0.013, +0.016]	yes
`collect_wood`	1.000	+0.0015	[+0.0010, +0.0020]	yes
`defeat_skeleton`	0.852	+0.009	[+0.004, +0.013]	yes
`defeat_zombie`	0.973	+0.019	[+0.016, +0.022]	yes
`eat_cow`	0.984	+0.012	[+0.009, +0.015]	yes
`make_stone_pickaxe`	1.000	+0.018	[+0.016, +0.019]	yes
`make_stone_sword`	1.000	+0.020	[+0.018, +0.022]	yes
`make_wood_pickaxe`	1.000	+0.008	[+0.007, +0.009]	yes
`make_wood_sword`	1.000	+0.042	[+0.039, +0.046]	yes
`place_furnace`	0.966	+0.015	[+0.012, +0.019]	yes
`place_plant`	1.000	−0.0000	[−0.0000, −0.0000]	no
`place_stone`	0.997	+0.017	[+0.015, +0.018]	yes
`place_table`	0.996	+0.016	[+0.013, +0.018]	yes
`wake_up`	0.916	+0.034	[+0.029, +0.041]	yes

Supplementary Table S5: **Sparse-autoencoder robustness sweep** on the block-1 summary activations (81,919 samples, 60 epochs), and single SAEs at other layers. FVU: fraction of variance unexplained on episode-disjoint validation. Detector features reappear as each variant's own top detector (mean matched decoder cosine 0.63).

Axis	Config (M, k, seed / layer)	FVU	dead features
width	1024, 16, 0	0.0445	0
(default)	2048, 16, 0	0.0440	0
width	4096, 16, 0	0.0441	0
sparsity	2048, 8, 0	0.0543	28
sparsity	2048, 32, 0	0.0363	0
seed	2048, 16, 1	0.0439	0
layer	frame_emb	0.267	—
layer	wm_input	0.0475	—
layer	block_2	0.0265	—
layer	block_3	0.214	—

Supplementary Table S6: **Empirical random-direction null.** Single-step matched-SAE ablation effect (gap 0) against an R=50 random-direction null; $z = (\text{matched} - \mu_{\text{null}})/\sigma_{\text{null}}$. The matched effect clears the 99th percentile for three of eight dominant concepts; honest negatives (e.g. `collect_iron`) sit at or below the null.

Concept	matched $\Delta \log P$	null $\mu \pm \sigma$	z	$> p_{99}$
`collect_coal`	+2.70	-0.02 ± 0.11	24.2	yes
`collect_sapling`	+1.73	$+0.02 \pm 0.17$	9.9	yes
`make_stone_pickaxe`	+0.61	-0.01 ± 0.20	3.1	yes
`place_plant`	+0.32	$+0.02 \pm 0.21$	1.4	no
`make_wood_pickaxe`	−0.16	-0.01 ± 0.15	−1.0	no
`place_table`	−0.04	-0.04 ± 0.20	0.0	no
`place_stone`	−0.11	-0.03 ± 0.19	−0.5	no
`collect_iron`	−0.39	-0.02 ± 0.16	−2.3	no

Supplementary Table S7: **Steering (sufficiency) inside imagination.** Adding the matched SAE direction ($h \mathrel{+}= \alpha\hat{d}$) from ordinary moments raises within-horizon detector-code probability only weakly; suppression (matched ablation) is shown for comparison. Mean induction gain +0.04 vs suppression drop +0.07 across 12 concepts (mean activate−suppress = −0.03): no activate-over-suppress asymmetry in this agent.

Concept	induce base	induce gain	suppress base	suppress drop
`collect_sapling`	0.00	+0.25	0.60	+0.20
`collect_stone`	0.15	+0.20	1.00	+0.00
`collect_iron`	0.05	+0.05	1.00	+0.15
`make_stone_pickaxe`	0.10	+0.05	0.80	+0.10
`place_stone`	0.05	+0.05	1.00	+0.00
`collect_coal`	0.05	−0.05	0.95	+0.25
`eat_cow`	0.15	−0.05	0.75	+0.00
`make_stone_sword`	0.10	+0.00	0.75	+0.30
`wake_up`	0.00	+0.00	0.30	−0.20

Supplementary Table S8: **Direct logit attribution** to the detector code, mean over 18 concepts (n=40 unlock moments each). $\overline{|\mathrm{det}|}$: mean absolute attribution to the detector code; $\overline{|\mathrm{det}| - |\mathrm{ctrl}|}$: detector-vs-control specificity; #sig: concepts whose detector attribution CI excludes 0. The SAE direction's attribution is $\sim 4\times$ the CAV's and $\sim 24\times$ a random direction's; the sign is concept-dependent (off-path `collect_iron` −1.56, `wake_up` +1.50).

Direction	$\overline{\lvert\mathrm{det}\rvert}$	$\overline{\lvert\mathrm{det}\rvert - \lvert\mathrm{ctrl}\rvert}$	#sig / 18
SAE feature	0.311	+0.073	9
CAV	0.083	+0.024	6
random	0.013	+0.000	2